NewsChain

A news app that starts from the outcome and rewinds time to reveal the full story.

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Contribution

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1. App Idea for Prototype

1.1 Description of the App

NewsChain is a narrative-driven mobile news application that helps users understand current events by starting from the present outcome and rewinding through past events to explain how and why the news happened. Rather than presenting an isolated headline, NewsChain presents news as a connected chain of events — each node in the chain is a distinct card showing a headline, source, date, and brief 2–3 sentence summary. Users can tap backwards through the chain to trace cause-and-effect relationships from today's outcome all the way to the origin event.

1.2 How Is It Novel?

Unlike traditional news apps (Apple News, Google News, BBC News) that prioritize the latest headlines in a chronological feed, NewsChain reverses the timeline of news consumption. By beginning with the outcome and guiding users backwards through relevant context, the app promotes deeper understanding and reduces surface-level, reactionary news reading. No existing mainstream news application presents news as a visual reverse-timeline chain of interconnected events with source attribution at every node.

1.3 Transformative Experience for the User

NewsChain creates four distinct shifts in the user experience:

- Grounded Perspective — Users see the present event as a single, clear moment of significance rather than an isolated headline.
- Sense of Discovery — Users experience curiosity and engagement as they uncover past events that led to today's news.
- Narrative-Driven Reading — News consumption becomes story-like, with cause and effect made explicit.
- Critical Thinking — By seeing full event chains, users develop a well-informed understanding of complex events and can better detect misinformation.

App Motto: "Starting with the outcome, trace back to the cause."

2. Background Research

2.1 Market Competitor Apps

The following apps represent the primary competitive landscape for NewsChain:

- Apple News — Aggregates personalized news in a clean feed; lacks historical story context or timeline navigation.
- Google News — AI-powered news aggregator organized by topic; stories appear in isolation without a reverse-timeline chain feature.
- BBC News App — Editorial news with high credibility; presents chronological article lists without cause-and-effect mapping.
- The New York Times App — Premium content with deep journalism; subscription-based, no visual story chain navigation.

2.2 Market Inspiration Apps

The following apps provided design inspiration and conceptual direction for NewsChain:

- Wikipedia — Demonstrates how complex events can be understood through linked historical context and references. Inspired the "chain" linking model in NewsChain.
- Notion / Knowledge Graph Tools — Show how information can be presented as connected nodes rather than linear feeds. Inspired the node-based card design in NewsChain.

3. PACT Framework

3.1 People

The primary users of NewsChain are:

- University and college students who need contextual understanding of current events for academic or civic engagement.
- Young professionals (ages 25–40) who follow industry news but have limited time to read multiple full articles.
- Casual news readers who feel overwhelmed or under-informed by conventional news feeds.
- Skeptical readers who want to verify source credibility before trusting and sharing a news story.

3.2 Activity

Users open NewsChain when they encounter a headline they want to understand more deeply. The core activity is: (1) viewing a trending news card on the home feed, (2) tapping into the story chain starting from the most recent outcome node, and (3) swiping or tapping backwards through historical event nodes to trace the full story. Secondary activities include bookmarking chains, sharing chains, and exploring topics via the search screen.

3.3 Context

The app is primarily used during commutes, short breaks, and study sessions. Users are often in mobile, distracted, or time-limited environments — on the SkyTrain, between classes, or on a lunch break. This context demands that content be short, scannable, and easy to navigate with one hand. The reverse-timeline chain format is well-suited to this context because each node delivers digestible value in under 30 seconds.

3.4 Technology

NewsChain is designed as a native mobile application (iOS and Android). Key technologies required include:

- Timeline-based navigation — swipe and tap gesture support for navigating event chains.
- Source aggregation — integration with news APIs to compile multi-source event nodes.
- Data visualization — visual chain rendering to connect event nodes.
- Personalization engine — topic preference matching to curate the home feed.
- Bookmarking and offline storage — for saving chains across sessions.

4. Value Proposition

NewsChain provides a new kind of reading experience for users who want to understand current events deeply, not just skim headlines. Compared to traditional news apps like Apple News and Google News, NewsChain starts with the outcome of an event and rewinds through past events to reveal cause-and-effect relationships explicitly.

This approach delivers three core value advantages:

- Depth without effort — Users gain full story context in a single, navigable chain rather than searching across five separate articles.
- Trust and transparency — Source attribution and credibility indicators on every node help users assess reliability at a glance.
- Engagement and discovery — The reverse-timeline format turns passive headline-skimming into active story exploration, fostering critical thinking and reducing misinformation susceptibility.

By combining timeline-based navigation, concise node summaries, and visualized event connections, NewsChain makes news understandable and easily consumable within the time constraints of modern life.

5. Requirements Gathering (User Study 1)

5.1 Purpose and Objectives

The objective of User Study 1 was to understand how people currently consume news and to identify pain points that NewsChain can address. Specifically, the study aimed to:

- Understand the news consumption habits and preferred platforms of the target users.
- Identify how often users feel confused or lacking context when encountering a breaking news headline.
- Determine user interest in a reverse-timeline news format that starts with a headline outcome and progressively reveals background context.
- Assess user preferences for news depth, from brief summaries to full historical chains of events.
- Understand what motivates or discourages users from reading full news articles to discover design priorities.
- Evaluate whether users would want personalization features such as topic preferences and saved story chains.

5.2 Choice of User Study and User Sample

User Study Type: An online survey was selected as the user study method. A survey allows for rapid, scalable data collection across a diverse group and is well-suited for gathering both quantitative data (frequency, ratings) and qualitative data (open-ended responses) using Google Forms.

Target Users: The target users for NewsChain are adults across a wide age range who regularly consume digital news, particularly through smartphones. This includes university and college students, young professionals, and casual news readers.

Dual Perspective: A single user type (general digital news readers) was used for this study, as NewsChain is primarily a consumer-facing news application without a content-publisher side.

Sample Size: 12 participants were surveyed. Participants included housemates, friends, and relatives.

5.3 Draft of the User Study Survey

Section A – Demographic Information

- Q1. What is your age range? (Under 18 / 18–24 / 25–34 / 35–44 / 45 or older)
- Q2. What is your occupation? (Student / Working Professional / Freelancer / Unemployed / Other)
- Q3. How many hours per day do you typically spend consuming news? (Less than 30 min / 30 min–1 hour / 1–2 hours / More than 2 hours)

Section B – Current News Habits

- Q4. Which platform(s) do you use most to follow news? (Social Media / News Apps / News Websites / TV/Radio / Word of mouth / Other)
- Q5. On a scale of 1–5, how often do you feel you lack background context when you encounter a breaking news headline? (1 = Never, 5 = Always)
- Q6. When you encounter a confusing news story, what do you typically do? (Search online / Ask someone / Ignore it / Skim multiple headlines / Other)

Section C – Interest in NewsChain's Concept

- Q7. How interested are you in a news format where stories start from the latest headline and you can tap backwards to see how the story developed over time? (1 = Not interested, 5 = Very interested)
- Q8. How important is it that a news app gives you the full historical context of a story? (1 = Not important, 5 = Very important)
- Q9. Would you prefer news stories presented as: Reverse timeline / Traditional chronological order / No preference
- Q10. On a scale of 1–5, how valuable would it be to see a visual chain connecting past events to a current headline? (1 = Not valuable, 5 = Very valuable)

Section D – Feature Preferences

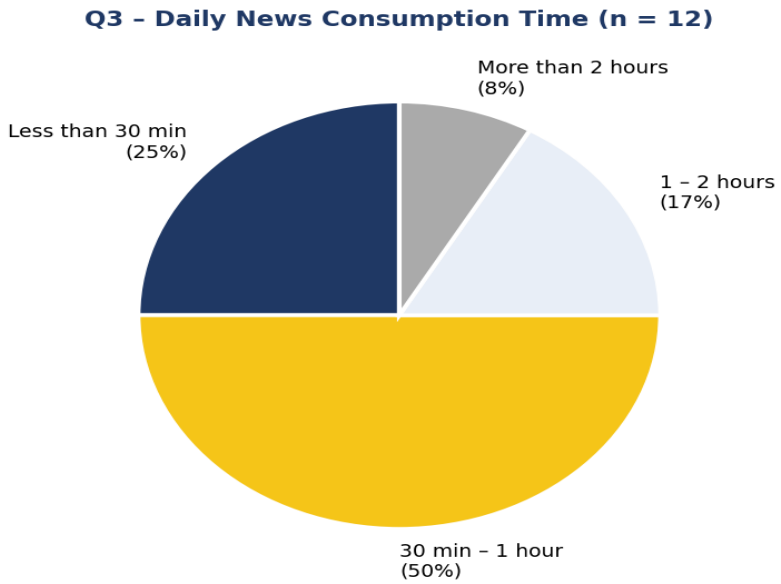
- Q11. Which features would you most want in a news app? (Bookmarking / Personalized feed / Offline reading / Story chain visualization / Share story chain / Dark mode / Audio summary)
- Q12. How important is it for the app to be available offline? (1 = Not important, 5 = Essential)

5.4 Data Visualization and Analysis

Q3 – Daily News Consumption Time (n = 12)

Time Spent	Responses	Percentage
Less than 30 min	3	25%
30 min – 1 hour	6	50%
1 – 2 hours	2	17%
More than 2 hours	1	8%

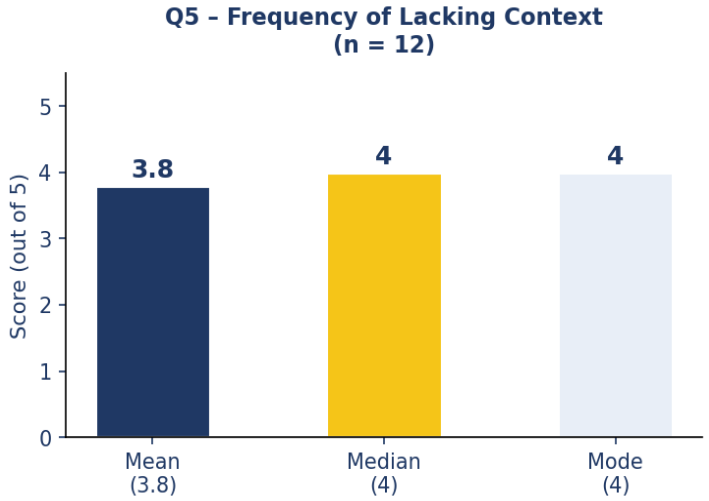
The majority of users (50%) spend 30 minutes to 1 hour per day on news consumption, indicating a moderate but consistent engagement level. This reinforces the need for concise, scannable content within NewsChain.



Q5 – Frequency of Lacking Context (Mean Score, n = 12)

Metric	Score (out of 5)
Mean	3.8
Median	4
Mode	4

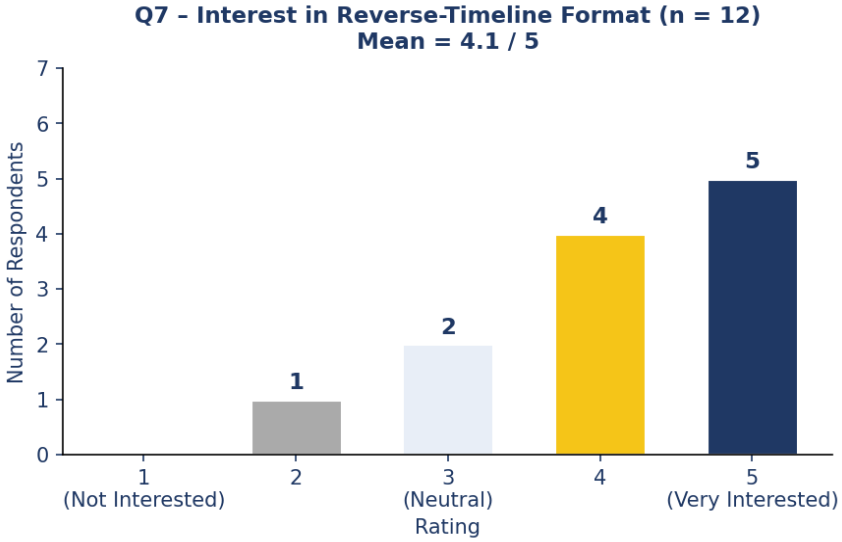
A mean score of 3.8 out of 5 indicates that users frequently feel they lack background context when reading breaking news. This strongly validates the core concept of NewsChain.



Q7 – Interest in Reverse-Timeline News Format (n = 12)

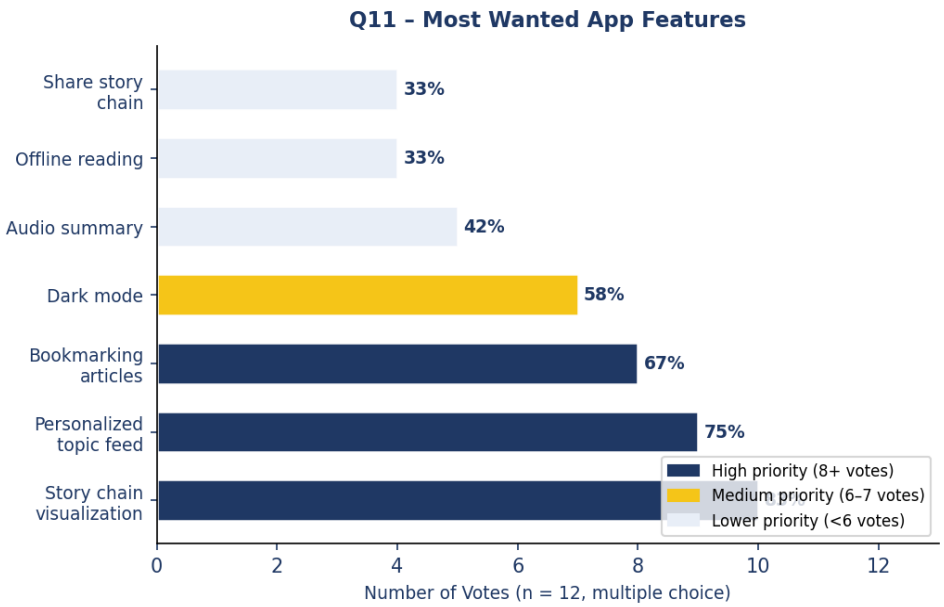
Rating	Responses	Percentage
1 – Not Interested	0	0%
2	1	8%
3 – Neutral	2	17%
4	4	33%
5 – Very Interested	5	42%

Mean: 4.1 / 5. A total of 75% of respondents rated their interest as 4 or 5, indicating strong demand for the reverse-timeline concept that is central to NewsChain.



Q11 – Most Wanted App Features (n = 12, multiple choice)

Feature	Votes	Percentage
Story chain visualization	10	83%
Personalized topic feed	9	75%
Bookmarking articles	8	67%
Dark mode	7	58%
Audio summary	5	42%
Offline reading	4	33%
Share story chain	4	33%



5.5 Insights from the Results

- Users frequently encounter news without sufficient context: A mean score of 3.8/5 on Q5 confirms that lack of background information is a widespread pain point, directly validating the NewsChain concept of presenting stories as historical chains.
- Strong interest in the reverse-timeline format: 75% of users rated their interest as 4 or 5, confirming the unique reverse-timeline format of NewsChain has genuine market appeal.
- Story chain visualization is the most desired feature: 83% of respondents selected it, reinforcing that the visual representation of a story's evolution is a key differentiator to build as a core feature.
- Personalization is highly valued: 75% want a personalized feed, meaning NewsChain should include topic preference settings to tailor which story chains appear for each user.
- Users are time-constrained: Most users spend 30–60 minutes per day on news, suggesting short, scannable chain summaries are essential before users dive into full chain exploration.
- Users cannot tell where they are within a story chain: Survey responses indicated that users had no way to know their current position within a chain while navigating. This insight led directly to the addition of an in-chain positional indicator (e.g., 'Event 2 of 4') displayed within the chain navigation view in the Item Detail screen, so users always know how deep into the chain they are.

6. Requirements Generation

6.1 Personas and Scenarios

Persona 1 – Maya Chen, the Overwhelmed Student

Name	Maya Chen
Age	21
Occupation	Science student at a large university
Tech Use	Heavy smartphone user; uses Instagram, Reddit, and TikTok for news
Frustration	Constantly sees breaking headlines that reference events she doesn't know about. Feels embarrassed in class discussions when she lacks context.
Goal	To quickly understand the background of a news story without hours of research

Scenario: Maya opens her phone between classes and sees a headline: 'Parliament passes emergency AI regulation bill.' She has no idea what led to this. She opens NewsChain, taps the headline, and is immediately presented with the most recent event node. She swipes backwards through a visual chain: the earlier Senate hearings, the AI incident that triggered public debate, and the original research paper from two years ago that started it all. In under 5 minutes, Maya understands the full story. She saves the chain to review later for a class essay.

Persona 2 – David Tanaka, the Busy Professional

Name	David Tanaka
Age	34
Occupation	Product Manager at a software company
Tech Use	Uses Apple News and LinkedIn for news; checks phone during commute
Frustration	News articles are long but shallow. He wants the gist of a story's full history fast, without reading five separate articles.
Goal	Stay informed on tech and business news efficiently during his 20-minute commute

Scenario: David is on the SkyTrain heading to work. He sees that a major tech company just announced a massive layoff. He opens NewsChain and taps the story. He is shown a concise chain: today's layoff announcement, the earnings report from last quarter, the pivot strategy announced 6 months ago, and the acquisition that started everything 2 years prior. He skims each node summary and feels fully caught up by the time he arrives at his stop. He shares the story chain link with a colleague via the app's share feature.

Persona 3 – Sandra Kowalski, the Sceptical News Reader

Name	Sandra Kowalski

Age	40
Occupation	High school teacher and parent of two
Tech Use	Moderate smartphone user; uses Facebook and Google News
Frustration	Distrusts sensationalist headlines; wants to see which sources are cited and whether a story has been consistent over time.
Goal	Find trustworthy, well-sourced news she can confidently discuss or share with students

Scenario: Sandra reads a controversial headline about a new school policy being debated in her province. She opens NewsChain and navigates the chain. She notices each node displays the source publication and a source-reliability badge. She can see that the story's framing changed significantly between outlets over the months, and she can tap through to read the original government document. Feeling confident in the sourced information, she decides to use the story chain as a classroom discussion resource, having bookmarked it the night before.

Persona 4 – Kenji Ohmura, the Casual News Scroller

Name	Kenji Ohmura
Age	25
Occupation	Barista and part-time freelance photographer
Tech Use	Heavy TikTok and Instagram user; occasionally reads news from social shares
Frustration	Finds traditional news apps boring and cluttered with too much text and unrelated ads.
Goal	Stay loosely informed on world events in a visually engaging, low-effort way

Scenario: Kenji is on his break at work and scrolls through NewsChain's home feed. The card-based layout with bold headline images catches his eye. He swipes through a visually-presented chain about a climate summit, appreciating the clean cards with icons representing each event node. He doesn't read every card fully but skims the visual timeline enough to feel informed. He taps the 'audio summary' button and listens to a 90-second recap of the full chain while cleaning equipment, completing his news catch-up effortlessly.

6.2 Requirements Generated from Persona and Scenario Analysis

From Maya Chen (Persona 1):

- The app must present a visual reverse-timeline chain for each news story.
- Each node in the chain must include a short summary (2–3 sentences) viewable without opening the full article.
- Users must be able to bookmark or save story chains for later reference.
- The app must allow users to swipe or tap backward through a chain naturally on mobile.

From David Tanaka (Persona 2):

- Chain node summaries must be scannable in under 30 seconds each.
- The app must provide a 'full chain summary' view that gives an overview of the entire story arc in one scroll.
- A share feature must allow users to send a story chain link to others via messaging apps, email, etc.
- The app must function well in low-connectivity environments such as during transit.

From Sandra Kowalski (Persona 3):

- Each node must display the source publication name and date clearly.
- The app should include a source credibility or reliability indicator on each node.
- Users must be able to tap through to read the original full article from any node.
- Bookmarks must be organized and persistently saved across app sessions.

From Kenji Ohmura (Persona 4):

- The home feed must use a card-based visual layout with compelling imagery.
- The app must support a dark mode for comfortable viewing in different lighting.
- An audio summary feature should allow users to listen to a 60–90 second recap of any story chain.
- The UI must be clean and uncluttered, with no intrusive advertisements interrupting story flow.
- Chain nodes should include visual icons or category tags (e.g., Politics, Tech, Climate) for quick orientation.

7. Synthesis of Requirements Gathering and Generation

7.1 Prioritized Requirements List

Must-Have Requirements (High Priority)

- Reverse-timeline story chain: The app must display news stories as a chain of linked events ordered from the most recent outcome backwards to the origin. This is the core differentiating feature of NewsChain.
- Node-based story display: Each event in the chain must be displayed as a distinct card/node with a headline, source, date, and a brief 2–3 sentence summary.
- Swipe-back navigation: Users must be able to navigate backwards through the chain using intuitive swipe or tap gestures, consistent with mobile UX conventions.
- Source attribution: Every node must clearly display the original news source name and publication date.
- Link to full article: Users must be able to tap any node to open the original full article, either in-app or in a browser.
- Bookmarking: Users must be able to save story chains to a personal reading list for future access.
- Home feed: A visually-engaging, card-based home screen must display trending and personalized story chains.
- Onboarding and topic selection: On first launch, users must be able to select their preferred news topics to personalize their feed.
- Chain event count indicator: Within the chain navigation view, a positional indicator (e.g., 'Event 2 of 4') must be shown so users always know where they are within the chain. This requirement was identified directly from User Study 1 data.

Want-to-Have Requirements (Medium Priority)

- Full chain summary view: A dedicated single-screen view that summarizes the entire story arc in one pass.
- Source credibility indicator: A visual badge or rating alongside each source to help users assess reliability.
- Dark mode: A full dark mode theme for the app interface to support comfortable reading at night or in low-light settings.
- Category and topic tags on nodes: Visual tags (e.g., 'Economy', 'Climate', 'Tech') to help users orient themselves within a chain.

Low Priority / Nice-to-Have Requirements

- Offline reading: Allow users to download story chains for offline reading during low-connectivity situations.
- Push notifications: Smart, minimal notifications for major developments in a user's followed story chains.
- Search functionality: Allow users to search for specific stories, topics, or keywords to find relevant chains.
- Chain comparison: Allow users to view how different outlets frame the same chain of events side by side.

- Audio summary: Users can listen to news without using their hands.

7.2 Conclusions from the Requirements Process

- The core NewsChain concept of reverse-timeline news is strongly validated by both user study data and persona analysis. It directly addresses the near-universal pain point of lacking context for breaking news.
- Visual design and UX simplicity are critical: cluttered UI is the most common reason users abandon news apps. Every screen in the Figma prototype was designed to be clean and card-centric.
- Source credibility and trust are underserved in current news apps. Building transparency into each chain node is a meaningful differentiator that will attract skeptical, educated users.
- Personalization is expected by modern news readers. The onboarding topic selection and personalized home feed address this directly.

8. Design

8.1 Design Patterns Chosen

- Card-Based Layout: All news items and story chain nodes are presented as cards with a consistent structure (image, category tag, headline, metadata). This pattern supports scannability and visual hierarchy.
- Bottom Navigation Bar: Persistent navigation between four main sections (Home, Explore, Bookmarks, Profile) using icon-based tabs. Standard in mobile applications; reduces navigation complexity.
- Reverse-Timeline Chain: The core interaction pattern where users start at the latest node and progress backwards in time by tapping or swiping. This is a novel application of a timeline pattern.
- Progress Tracking: The Bookmark screen doubles as a reading progress tracker, showing percentage completion of each saved chain. This encourages re-engagement.
- Gamification: The Profile screen includes streak tracking, achievement badges, and saved/completed counts. This encourages daily news habits.

8.2 Rationale for Design Choices

- Yellow and dark color palette: The bold yellow accent creates high visibility and energetic branding, consistent with an app promoting active news engagement. Dark text on yellow cards ensures strong contrast and readability.
- Category tags as colored chips: Color-coded category tags (e.g., green for Environment, blue for Technology) allow users to quickly identify topic domains within a dense card list.
- Credibility score on nodes: Displaying a credibility percentage (e.g., 90% Credibility) on the ItemDetail screen addresses Sandra Kowalski's need for source trust, validated by the user study findings.
- Minimal navigation depth: All screens are accessible within one tap from the bottom navigation bar, reducing cognitive load for time-constrained users like David Tanaka.

8.3 Overall Workflow on Figma

The Figma prototype (available at: https://www.figma.com/design/f8frhn1ZJTwjYEX1t53uni/) consists of eight primary screens: Home, Explore, Bookmark (Reading Path), Profile, Item Detail, Chain Navigation, News Detail, and Credibility. Five screens (Home, Explore, Bookmark, Profile, Item Detail) are connected via the persistent Bottom Navigation Bar. The remaining three screens (Chain Navigation, News Detail, Credibility) are accessed through in-screen interactions. The full navigation flow is as follows:

Home Screen

- Home → Explore Screen: Via the Search icon in the nav bar.
- Home → Chain Navigation: Via tapping the 'How to interact' text.
- Home → Item Detail: Via tapping a news card.
- Home → Bookmark Screen: Via the Bookmark icon in the nav bar.

- Home → Profile Screen: Via the Profile icon in the nav bar.

Explore Screen

- Explore → Home Screen: Via the Home icon in the nav bar.
- Explore → Bookmark Screen: Via the Bookmark icon in the nav bar.
- Explore → Profile Screen: Via the Profile icon in the nav bar.
- Explore → Item Detail Screen: Via tapping a news card.

Bookmark (Reading Path) Screen

- Bookmark → Item Detail Screen: Via tapping a news card.
- Bookmark → Home Screen: Via the Home icon in the nav bar.
- Bookmark → Explore Screen: Via the Explore icon in the nav bar.
- Bookmark → Profile Screen: Via the Profile icon in the nav bar.

Profile Screen

- Profile → Home Screen: Via the Home icon in the nav bar.
- Profile → Explore Screen: Via the Explore icon in the nav bar.
- Profile → Bookmark Screen: Via the Bookmark icon in the nav bar.

Item Detail Screen

- Item Detail → News Detail: Via tapping a news card.
- Item Detail → Home Screen: Via the Home icon in the nav bar.
- Item Detail → Explore Screen: Via the Explore icon in the nav bar.
- Item Detail → Bookmark Screen: Via the Bookmark icon in the nav bar.
- Item Detail → Profile Screen: Via the Profile icon in the nav bar.
- Item Detail / News Detail → Credibility Screen: Via 'View Detail' button.

Chain Navigation Screen

- Chain Navigation → Home Screen: Via the door icon.

News Detail Screen

- News Detail → Item Detail: Via tapping the left direction arrow.

Credibility Screen

- Credibility → News Detail: Via tapping the left direction arrow.

8.4 Screenshots of All Screens

Screen 1: Home Feed

The Home screen displays recommended news story chains as large cards with a headline image, category tag (e.g., Environment, Technology), trending indicator, read time, and date range showing the chain's time span. The NewsChain logo and brand name appear in the yellow header. A bottom navigation bar provides persistent access to all sections. A 'How to interact' text link on the Home screen leads users to the Chain Navigation tutorial screen.

[Figma Screenshot – Home Screen:
https://www.figma.com/design/f8frhn1ZJTwjYEX1t53uni/?node-id=48-33]

Screen 2: Explore / Search

The Explore screen provides a search bar for finding news by topic, event, or keyword. Below the search bar, a horizontal scrollable row of category filter chips (All, Environment, Technology, Economy, etc.) allows quick topic filtering. A 'Trending Topics' section displays popular topic tags, followed by a list of news cards matching the current filter. This screen addresses the requirement for search functionality and topic-based discovery.

[Figma Screenshot – Explore Screen:
https://www.figma.com/design/f8frhn1ZJTwjYEX1t53uni/?node-id=48-35]

Screen 3: Reading Path (Bookmarks)

The Bookmark screen is branded as 'Your Reading Path' and doubles as a reading progress tracker. It displays three summary statistics: Saved chains (2), Completed chains (0), and overall Progress percentage (38%). A 'Continue Reading' section shows partially-read chains with progress bars indicating completion percentage. A 'Recommended For You' section appears below. This screen supports the requirements for persistent bookmarking and reading progress tracking.

[Figma Screenshot – Bookmark/Reading Path Screen:
https://www.figma.com/design/f8frhn1ZJTwjYEX1t53uni/?node-id=48-66]

Screen 4: Profile

The Profile screen shows the user's identity (username: Reader, label: Critical Thinker), a streak count (7 days), and saved/done/events statistics. A 'Recent Achievements' section displays earned badges (7 Days Streak, First Chain Completed). A 'Your Interests' section displays selected topic tags that personalize the home feed. This screen supports the gamification and personalization requirements.

[Figma Screenshot – Profile Screen:
https://www.figma.com/design/f8frhn1ZJTwjYEX1t53uni/?node-id=48-97]

Screen 5: Item Detail (Story Chain View)

The Item Detail screen is the core screen of NewsChain. It presents the most recent outcome

node prominently at the top with a full-width hero image, category tag, headline, read time, event count, and a trending indicator. A highlighted 'The Outcome Today' section introduces the reverse-timeline experience. Below it, the first chain node card is displayed with the date, headline, and a credibility score (e.g., 90% Credibility), with a 'View Detail >' link to the Credibility screen. Bookmark and Share buttons appear in the top-right corner. Users can tap a node card to go to the News Detail screen, or use the nav bar to navigate elsewhere.

[Figma Screenshot – Item Detail Screen:
https://www.figma.com/design/f8frhn1ZJTwjYEX1t53uni/?node-id=146-808]

Screen 6: Chain Navigation

The Chain Navigation screen is an onboarding/tutorial screen accessible from the Home screen via the 'How to interact' text link. It introduces users to the reverse-timeline chain concept, explaining how to navigate backwards through event nodes. Users can return to the Home screen via the door icon. This screen directly addresses the onboarding gap identified in the User Study 2 feedback.

[Figma Screenshot – Chain Navigation Screen:
https://www.figma.com/design/f8frhn1ZJTwjYEX1t53uni/]

Screen 7: News Detail

The News Detail screen displays the full content of an individual event node within a story chain. Users arrive here by tapping a news card from the Item Detail screen. The screen shows the full article summary, source information, and the credibility score with a 'View Detail >' button leading to the Credibility screen. A left direction arrow at the top allows users to navigate back to the Item Detail screen.

[Figma Screenshot – News Detail Screen:
https://www.figma.com/design/f8frhn1ZJTwjYEX1t53uni/]

Screen 8: Credibility

The Credibility screen provides a detailed breakdown of the source credibility assessment for a specific news node. It is accessible from both the Item Detail and News Detail screens via the 'View Detail >' button. The screen displays the credibility score, source publication details, and supporting evidence for the rating. Users can return to the News Detail screen via the left direction arrow.

[Figma Screenshot – Credibility Screen:
https://www.figma.com/design/f8frhn1ZJTwjYEX1t53uni/]

9. Designer-Based Prototype Evaluation

9.1 User Journey Map

The User Journey Map below traces the primary path of a target user — represented by Persona 2 (David Tanaka, Busy Professional) — through the core NewsChain experience.

Phase	Awareness	Exploration	Engagement	Deep Reading	Return Visit
Touchpoint	App store / social share	Home screen, onboarding	Story chain / Item Detail	Node tap, source view	Bookmark / Reading Path
User Action	Downloads app, sees intro	Selects topics, browses feed	Taps news card, enters chain	Swipes back through nodes, taps source	Re-opens Reading Path, continues
Emotion	Curious, hopeful	Oriented, interested	Engaged, discovering	Informed, satisfied	Habitual, accomplished
Pain Point	–	Uncertain how chain works	Chain depth unclear at first	Wants to verify credibility	Needs reminder to return

10. User-Based Prototype Evaluation (User Study 2)

10.1 Purpose and Objectives

The objective of User Study 2 was to evaluate the usability and clarity of the NewsChain Figma prototype with real users. Specifically, the study aimed to:

- Determine whether users understand the reverse-timeline chain concept on first encounter.
- Identify any navigation pain points or confusion within the prototype flows.
- Assess the clarity of the source credibility indicator and bookmarking feature.
- Gather qualitative feedback on the visual design, card layout, and information density.
- Validate whether the prototype meets the core user needs identified in User Study 1.

10.2 Choice of User Study and User Sample

User Study Type: An in-person think-aloud usability test combined with a short post-task survey. Participants were asked to complete three tasks using the Figma prototype while verbalizing their thoughts. The post-task survey used a 1–5 Likert scale to capture quantitative ratings, followed by open-ended questions for qualitative feedback.

Target Users: Same as User Study 1 — general digital news readers aged 18–45. Participants were recruited from the same social circle as Study 1 (housemates and relatives). A single user perspective was used (general news readers) consistent with Study 1.

Sample Size: 5 participants completed the prototype evaluation. Manish, Darhist, and Abby are housemates; Riku and Toshi are relatives.

10.3 Draft of the User Study Survey

The full survey is included in Appendix 14.5. Participants completed the following think-aloud tasks using the Figma prototype, then answered the post-task survey below.

Think-Aloud Tasks (administered before the survey)

- Task 1: Please find a technology news story on the Home screen and open its story chain.
- Task 2: Once inside the story chain, navigate backwards through at least two event nodes.
- Task 3: Bookmark the story chain, then locate your saved chain in the Reading Path screen.

Section A – Participant Information

- Q1. What is your age range? (Under 18 / 18–24 / 25–34 / 35–44 / 45 or older)
- Q2. How often do you currently use news apps? (Daily / A few times a week / Occasionally / Rarely / Never)

Section B – App Navigation Experience

- Q3. How easy was it to find a news story you were interested in on the Home screen? (1 = Very Difficult, 5 = Very Easy)
- Q4. How easy was it to navigate backwards through the story chain events? (1 = Very Difficult, 5 = Very Easy)

Section C – Prototype Usability Ratings

- Q5. How clearly did the app communicate the concept of starting from the latest news and going back in time? (1 = Not Clear at All, 5 = Very Clear)
- Q6. How useful was the credibility score shown on each story node? (1 = Not Useful, 5 = Very Useful)
- Q7. Overall, how satisfied are you with the NewsChain app prototype? (1 = Very Dissatisfied, 5 = Very Satisfied)

Section D – Feature Feedback

- Q8. What did you like most about the app? (Open-ended)
- Q9. What would you change or improve about the app? (Open-ended)

10.4 Data Visualization and Analysis

Sample size: 5 participants. Sections B and C of the survey used a 1–5 Likert scale. The table below shows individual participant responses for each rated question.

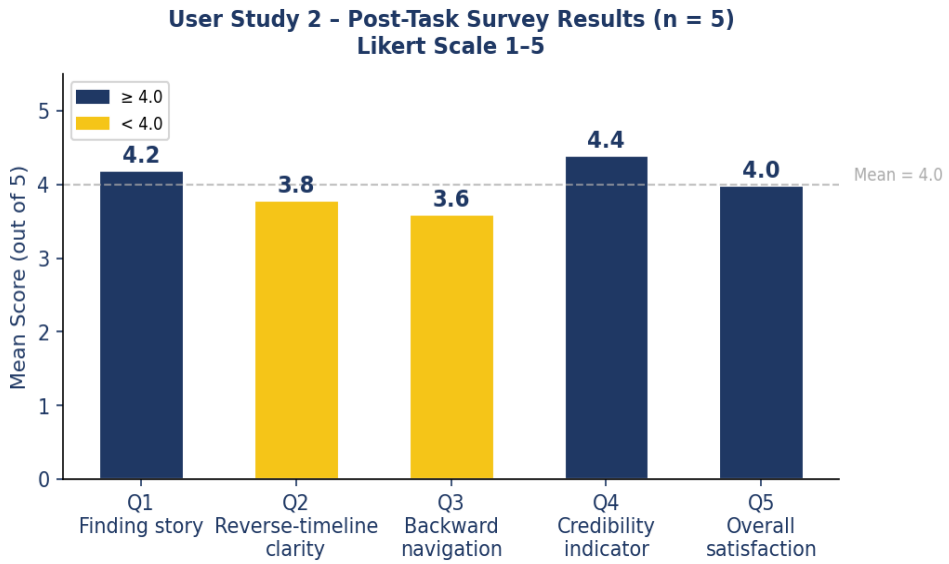
Individual Participant Responses

Participant	Q3	Q4	Q5	Q6	Q7
Manish (housemate, 22)	4	3	3	5	4
Riku (relative, 20)	4	4	3	4	4
Darhist (housemate, 31)	4	4	4	5	4
Abby (housemate, 28)	4	4	4	5	4
Toshi (relative, 35)	5	4	4	3	4
Mean	**4.2**	**3.8**	**3.6**	**4.4**	**4.0**
Median	**4**	**4**	**4**	**5**	**4**

Summary Statistics

Survey Question	Mean	Median
Q3 – Ease of finding a story on Home screen	4.2 / 5	4
Q4 – Ease of backward chain navigation	3.6 / 5	4
Q5 – Clarity of reverse-timeline concept	3.8 / 5	4

Q6 – Usefulness of credibility indicator	4.4 / 5	5
Q7 – Overall satisfaction with the prototype	4.0 / 5	4



Open-Ended Responses (Q8 & Q9)

Q8 – What did you like most about the app?

- "The card layout on the home screen is really clean and easy to browse." (Manish)
- "I loved the idea of seeing where the story started — it felt like reading a mystery backwards." (Riku)
- "The credibility score was really helpful. I always wonder if news is trustworthy." (Darhist)
- "The yellow design is eye-catching and different from other news apps." (Abby)
- "Being able to bookmark and track reading progress is a great feature." (Toshi)

Q9 – What would you change or improve?

- "I wasn't sure at first which way to swipe to go back in the chain. A hint would help." (Manish)
- "I didn't realize I was supposed to go backward — a small tutorial at the start would be useful." (Riku)
- "The number of events in the chain wasn't clear. Something like 'Event 2 of 5' would help." (Darhist)
- "Would be nice to have a button that says 'Previous Event' rather than just swiping." (Abby)
- "Nothing major — maybe just clearer labels on the navigation bar icons." (Toshi)

10.5 Insights from the Results

- The home feed card layout is effective: All participants found the Home screen visually engaging and easy to navigate. Q1 achieved the second-highest mean (4.2/5), confirming the card-based design is well-received.
- Navigation onboarding is the most critical gap — and was addressed: Q2 (3.8/5) and Q3 (3.6/5) were the two lowest-scoring items. Manish and Riku both verbalized that the reverse-timeline direction was not intuitive on first use, and Abby requested a visible 'Previous Event' button. This direct user feedback led to the addition of the Chain Navigation screen in the Figma prototype — a dedicated onboarding screen accessible from the Home screen that explains how the chain works before users dive in.
- Chain event count indicator validated and implemented: Darhist noted 'The number of events in the chain wasn't clear — something like Event 2 of 5 would help.' This directly validates the in-chain positional indicator (Event 2 of 4) added in response to User Study 1. The indicator was confirmed necessary by User Study 2 feedback and is implemented in the current Figma prototype.
- The credibility indicator is highly valued: Q4 achieved the highest mean (4.4/5) and the highest median (5/5). This confirms source transparency is a key differentiator and must be maintained as a core feature.
- Overall satisfaction is positive: Q5 mean of 4.0/5 with a consistent rating of 4 across all five participants indicates the prototype successfully meets the core needs identified in User Study 1.
- Bookmarking is intuitive: All participants successfully completed Task 3 without difficulty, validating the Reading Path screen design.

11. Changes to Design Based on Prototype Evaluation

11.1 Changes Needed Based on Feedback

- Improve backward navigation affordance by adding a visible 'Previous Event' button or directional arrow indicator below each node. (Feedback from Abby)
- Increase tap target size for the Bookmark button on each card to reduce accidental misses on smaller screens.
- Add a subtle animation on the chain node transition to reinforce the backward direction of travel.

11.2 Changes Made After Feedback

The following two changes were made directly in response to user feedback from User Study 1 and User Study 2. Other design features visible in the prototype (category tags, credibility badge, Reading Path progress bars, '4 events' card label) were original design decisions from the initial design phase based on persona analysis, not from prototype evaluation feedback.

In response to User Study 1

- In-chain positional indicator added: User Study 1 revealed that users had no way to track their position within a chain while navigating it. In direct response, an 'Event 2 of 4' style positional indicator was added within the chain navigation view in the Item Detail screen. This tells users exactly where they are in the chain at all times. The indicator is implemented in the current Figma prototype.

In response to User Study 2

- Chain Navigation (Onboarding) screen added: Manish and Riku reported that the reverse-timeline direction was not intuitive, and Abby requested a visible navigation cue. In direct response, a dedicated Chain Navigation screen was added to the prototype, accessible from the Home screen via 'How to interact'. It explains the chain concept before users begin navigating. This screen is implemented in the current Figma prototype.

11.3 Changes Not Yet Implemented (Major Revisions)

- Previous Event button: A persistent 'Previous Event' button on the News Detail screen would allow users to navigate back without relying solely on swipe gestures. This requires UI restructuring and is deferred to the next iteration. (Feedback from Abby)
- Onboarding animation: A polished, animated onboarding flow explaining the reverse-timeline concept requires significant design effort. The current Chain Navigation screen is a partial solution, but a fully animated flow is deferred to the next iteration.
- Audio summary feature: A 60–90 second AI-generated audio recap requires backend AI integration, beyond the current prototype scope.
- Offline reading mode: Caching full story chains for offline access requires local database storage and is not feasible in the current Figma prototype stage.

- Push notifications: Smart notifications for chain updates require backend notification services, deferred to production implementation.

12. Conclusion

NewsChain successfully demonstrates a novel and validated approach to mobile news consumption. By reversing the traditional chronological news feed and presenting news as visual chains of interconnected events, the app addresses a genuine and widespread user pain point: the lack of historical context when encountering breaking news headlines.

The two-phase research process — an initial user needs survey (n=12) followed by persona-driven requirements generation and prototype evaluation (n=5) — consistently validated the core product concept and directly drove design decisions. 75% of initial survey respondents rated their interest in the reverse-timeline format at 4 or 5 out of 5. Two clear research-to-design chains emerged: User Study 1 identified that users cannot track their position within a story chain, which led to adding an in-chain positional indicator ('Event 2 of 4') to the chain navigation view in Item Detail; and User Study 2 identified that the reverse-timeline navigation is not self-evident to first-time users, which led to the addition of the Chain Navigation onboarding screen accessible from the Home screen. These changes demonstrate that the prototype evaluation process functioned as intended — surfacing real usability issues and driving concrete, traceable design improvements.

The primary challenge identified is the onboarding experience: the reverse-timeline interaction model is novel and requires clear introduction for first-time users. This is the highest-priority design improvement for the next iteration.

The eight screens designed in Figma — Home, Explore, Reading Path, Profile, Item Detail, Chain Navigation, News Detail, and Credibility — cover the full core user journey from discovery to deep reading to source verification and return visits. The implementation of the Bottom Navigation Bar and Card List components (completed in Progress Report #1) establishes the foundational UI architecture for the production build.

NewsChain has strong market differentiation from existing news apps and a clear value proposition grounded in user research. With continued iteration on the onboarding experience and chain navigation affordances, the app is positioned to deliver a genuinely transformative news reading experience.

13. AI Use Section

13.1 Table of AI Tools and Specific Use

AI Tool Name	Version / Account Type	Specific Feature Used
ChatGPT	GPT-4o mini, Free account	Background research on competitor apps; example questionnaire generation; identifying key challenges in user requirements
Claude (Anthropic)	Claude Sonnet 4.6, claude.ai Free	Formatting Word document based on questionnaire results; Final Report writing and document structure

13.2 Value Addition

14. Appendix

14.1 Prompt History – All AI Tool Interactions

ChatGPT (GPT-4o mini) – February 1, 2026

Prompt: "Explain Background Research: competitor apps and inspiration apps for NewsChain."

Use: Used to compile an initial list of competitor apps (Apple News, Google News, Wikipedia, Vox) and their influence on the NewsChain concept. All content was reviewed, filtered, and rewritten by the student.

ChatGPT (GPT-4o mini) – March 2, 2026

Prompt 1: "Can you tell me where I should improve my questionnaire as user requirement"

Use: Reviewed the draft questionnaire for clarity improvements; student incorporated selected suggestions.

Prompt 2: "Can you tell me example of questionnaire based on the application concept"

Use: Generated example survey questions for a news application; student adapted and filtered these for relevance to NewsChain.

ChatGPT (GPT-4o mini) – March 9, 2026

Prompt: "What is main challenging point from user requirement"

Use: Used to identify common challenges in user requirements gathering. Output was reviewed and interpreted by the student in the context of NewsChain's specific findings.

Claude (Anthropic) – March 9, 2026

Prompt: "Can you format word file based on my questionnaire results?"

Use: Used to format the Requirements Document based on the student's questionnaire data

and written content. All content was written by the student; Claude provided structural formatting.

Claude (Anthropic) – April 11 2026

Prompt: “please analyze user study data and format word file?”

Use: Document generation assistance. All research, design, user study data, personas, requirements, and design decisions are the student's original work.

14.5 User Study 2 – Full Survey Draft (Prototype Evaluation)

The following is the complete survey administered to 5 participants (Manish, Darhist, Abby — housemates; Riku, Toshi — relatives) after completing the guided tasks on the Figma prototype. The questionnaire was administered in person.

Think-Aloud Tasks (Pre-Survey)

- Task 1: Please find a technology news story on the Home screen and open its story chain.
- Task 2: Once inside the story chain, navigate backwards through at least two event nodes.
- Task 3: Bookmark the story chain, then locate your saved chain in the Reading Path screen.

Section A – Participant Information

- Q1. What is your age range? (Under 18 / 18–24 / 25–34 / 35–44 / 45 or older)
- Q2. How often do you currently use news apps? (Daily / A few times a week / Occasionally / Rarely / Never)

Section B – App Navigation Experience

- Q3. How easy was it to find a news story you were interested in on the Home screen? (1 = Very Difficult, 5 = Very Easy)
- Q4. How easy was it to navigate backwards through the story chain events? (1 = Very Difficult, 5 = Very Easy)

Section C – Prototype Usability Ratings

- Q5. How clearly did the app communicate the concept of starting from the latest news and going back in time? (1 = Not Clear at All, 5 = Very Clear)
- Q6. How useful was the credibility score shown on each story node? (1 = Not Useful at All, 5 = Very Useful)
- Q7. Overall, how satisfied are you with the NewsChain app prototype? (1 = Very Dissatisfied, 5 = Very Satisfied)

Section D – Feature Feedback

- Q8. What did you like most about the app? (Free response)
- Q9. What would you change or improve about the app? (Free response)

Raw Data Collected (Sections B & C)

Participant	Q3	Q4	Q5	Q6	Q7
Manish (housemate, 22)	4	3	3	5	4
Riku (relative, 20)	4	4	3	4	4
Darhist (housemate, 31)	4	4	4	5	4
Abby (housemate, 28)	4	4	4	5	4
Toshi (relative, 48)	5	4	4	3	4
Mean	**4.2**	**3.8**	**3.6**	**4.4**	**4.0**
Median	**4**	**4**	**4**	**5**	**4**

14.2 Figma Prototype Link

Full interactive prototype: https://www.figma.com/design/f8frhn1ZJTwjYEX1t53uni/

14.3 Work Date / Hours Log

Date	Hours	Description
Feb 1, 2026	6.0	Writing summary of production proposal
Feb 17, 2026	3.0	Making Figma app prototype
Feb 17, 2026	2.0	Recording video, writing report
Mar 2, 2026	1.0	Generated requirements list from personas; organized into Must-Have / Want-to-Have categories based on Figma prototype
Mar 4, 2026	1.0	Posting Google questionnaire and discussing app concept with housemates and friends
Mar 9, 2026	1.5	Wrote synthesis section, conclusions, and AI use section; cross-checked requirements against Figma prototype
Mar 23, 2026	1.0	Implemented the Bottom Navigation Bar component for the NewsChain app
Mar 23, 2026	1.0	Implemented the Card List component for the home feed screen
Apr 2026	3.0	Prototype evaluation (User Study 2); writing Final Report

Total	**19.5**	

14.4 References

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